ZS-U5

Quantum Gravity Bridge: Z-Spin Structural Correspondence, Z-Telomere Collapse, and the Topological Life-Cycle

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Theme: Early Universe [ZS-U] | Paper 5 of 8

Verification: 34/34 PASS | Zero New Fit Parameters

§0. Abstract

We establish a structural correspondence between the Z-Spin Q=11 slot register and the 4-valent quantum tetrahedron of loop quantum gravity. The Q=11 register with sector decomposition (Z,X,Y) = (2,3,6) is derived from polyhedral geometry (ZS-F2/F5 v1.0), not from SU(2) recoupling theory. The correspondence is that both structures are organized by the same tetrahedral symmetry group T_d, sharing the group-theoretic quantities $|T_d| = 24$, $|Stab_{T_d}(v)| = 6$, and $|I_h/T_d| = 5$ that determine all Z-Spin timescales. [HYPOTHESIS]

The Z-Telomere mechanism introduces a dynamical phase-drift process from the Z-Spin action's non-minimal coupling $(1+A\varepsilon^2)R$. The phase drift $\delta\varphi = A$ is DERIVED-under-P6 via the Regge-Holonomy framework (§5.2: $\delta\varphi_{cell} = A \cdot I_{cell}$) combined with the corrected κ census and ZS-F5 v1.0 P6 bridge. The key derivation chain: ZS-F5 v1.0 $\kappa_{disrupted} \in \{0,4,6\}$ + P6 bridge ($\kappa \leq r = 4$) selects 3 diagonal involutions with $\kappa = 4$, yielding $I_{cell} = \kappa/r = 4/4 = 1$, hence $\delta\varphi_{cell} = A$ exactly.

Additionally, the complete one-loop RG trajectory for the quartic self-coupling is derived: $\beta_\lambda = (3/16\pi^2)(\lambda - 6A^2)(\lambda - 2A^2)$, yielding the IR-stable fixed point $\lambda_{vac} = 2A^2 = 0.01283$ [DERIVED-CONDITIONAL], which determines the radial mass $m_\rho = 2A \times M_P$ and cell size $\ell_c = \ell_P/(A\sqrt{2}) = 8.83\ \ell_P$.

Scope Note: ZS-U5 covers quantum gravity and topological structure exclusively. Post-inflationary thermalization belongs to ZS-U2 v1.0.

Keywords: quantum gravity, loop quantum gravity, Z-Spin structural correspondence, quantum tetrahedron, Immirzi parameter, Z-Telomere, Regge calculus, RG fixed point, topological life-cycle

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem with complete proof.
STANDARD	Established result in LQG/QFT textbooks.
DERIVED	Quantitative consequence from Z-Spin axioms + standard physics.
DERIVED-under-P6	Derived conditional on ZS-F5 v1.0 Proposition P6.
DERIVED-CONDITIONAL	Derived conditional on specified β-function form.
HYPOTHESIS	Conjectured with explicit falsification protocol.
NON-CLAIM	Explicitly excluded from this paper's scope.

§1. Locked Inputs

Table 1. Locked constants.

Constant	Value	Source
A	35/437 = 0.080092	ZS-F2 v1.0 [LOCKED]
Q	11	ZS-F2 v1.0 [LOCKED]
(Z,X,Y)	(2,3,6)	ZS-F2 v1.0 [LOCKED]
κ	4	ZS-F5 v1.0 P6 [LOCKED]
r (rank)	4	SU(2) recoupling [STANDARD]
γ_LQG	$\ln 2/(\pi\sqrt{3}) \approx 0.1274$	LQG [STANDARD]
λ_vac	$2A^2 = 0.01283$	This paper §8 [DERIVED-CONDITIONAL]

§2. Structural Correspondence with LQG Quantum Tetrahedron

2.1 Standard SU(2) Result

The 4-valent j=1/2 quantum tetrahedron in loop quantum gravity has the standard SU(2) Clebsch-Gordan decomposition:

$$(1/2)^{\otimes 4} = 2\cdot(j=0) \oplus 3\cdot(j=1) \oplus 1\cdot(j=2) \qquad (1)$$

Total Hilbert space dimension: $2^4 = 16$. Gauge-invariant intertwiner space dimension: 2 (multiplicity of j=0). These are standard SU(2) recoupling results.

[STATUS: STANDARD] Textbook SU(2) Clebsch-Gordan decomposition.

2.2 Z-Spin Register (Independent Origin)

The Z-Spin Q=11 slot register with (Z,X,Y) = (2,3,6) is derived from polyhedral geometry in ZS-F2 v1.0 and ZS-F5 v1.0, not from SU(2) recoupling. The sector dimensions are: Z=2 (parity sectors from tetrahedral self-duality), X=3 (spatial channels from cube-octahedron duality), Y=6 (gauge modes from dodecahedron-icosahedron duality).

$$Q = Z + X + Y = 2 + 3 + 6 = 11, \quad Z \times X \times Y = 36 = 6^2 \quad (2)$$

[STATUS: DERIVED] From ZS-F2/F5 v1.0 polyhedral geometry. LOCKED.

2.3 Structural Correspondence [HYPOTHESIS]

Both the LQG quantum tetrahedron and the Z-Spin register are organized by the tetrahedral symmetry group T_d. The correspondence shares the following group-theoretic quantities, all of which are rigorously proven:

(a) |T_d| = 24 (tetrahedral symmetry group order). (b) |Stab_{T_d}(v)| = 6 = $|D_3|$ (vertex stabilizer). (c) |I_h/T_d| = 120/24 = 5 (icosahedral coset dimension). (d) Regge deficit angles on tetrahedral cells (DERIVED-under-P6).

Critical distinction: The group-theoretic quantities (|T_d|, |Stab|, |I_h/T_d|) that determine all Z-Spin timescales (§5, §6) are rigorously proven and do NOT depend on the Q=11 correspondence. The correspondence itself — that the Z-Spin register's polyhedral origin and LQG's quantum tetrahedron reflect the same underlying geometric structure — remains a hypothesis pending a derivation connecting polyhedral impedance to spin-network state counting.

Explicit non-identity: Q = 11 ≠ dim(intertwiner) = 2 ≠ dim(Hilbert) = 16. The Z-Spin register counts sector slots, not quantum states. The number 11 arises from polyhedral sector decomposition (ZS-F2 v1.0), not from SU(2) recoupling algebra.

[STATUS: HYPOTHESIS] Structural correspondence. Group-theoretic inputs are PROVEN independently. The interpretive bridge is HYPOTHESIS.

§3. Black Hole Entropy and the ln 2 Origin

The standard LQG Bekenstein-Hawking entropy calculation yields S_BH = A_horizon/(4G) when the Immirzi parameter takes $\gamma = \ln 2/(\pi\sqrt{3})$. The ln 2 factor has a physical origin in Z-Spin: the Z=2 sector contributes a Z_2 parity entropy of exactly ln 2 per quantum tetrahedron face. The Z_2 seam automorphism (J) of the Z-Spin framework maps directly to the Z_2 parity symmetry that constrains the spin-network state counting. This is a structural interpretation of the standard LQG result, not a new derivation.

[STATUS: DERIVED] Structural interpretation of standard LQG result.

§4. Immirzi Parameter Connection

The Immirzi parameter $\gamma_LQG = \ln 2/(\pi\sqrt{3})$ connects to Z-Spin through two independent paths:

Path A (Z-parity): The Z=2 sector contributes binary entropy ln 2 per face. The $\pi\sqrt{3}$ denominator arises from the geometric factor of the equilateral tetrahedron (area = $\sqrt{3}/4 \times \text{side}^2$, with π from the Barbero-Immirzi quantization).

Path B (Thermal): The Tolman-Oppenheimer temperature at the Z-anchor yields $T_BH \propto 1/\gamma$, connecting gravitational thermodynamics to the microscopic Z_2 structure.

[STATUS: DERIVED] Two independent structural paths. No numerical fitting.

§5. Z-Telomere Collapse Mechanism

5.1 Phase Drift from Non-Minimal Coupling

The non-minimal coupling $(1+A\varepsilon^2)R$ generates a phase drift per oscillation cycle near topological defects: $\delta\varphi = A = 35/437 \approx 0.0801$ radians per cycle. This was upgraded from HYPOTHESIS to DERIVED-under-P6 via the Regge-Holonomy framework.

5.2 Regge-Holonomy Framework

The deficit angle per primitive Regge cell is: $\delta\varphi_cell = A \cdot I_cell$, where $I_cell = \kappa/r$. The corrected dual-κ census distinguishes $\kappa_signflip$ from $\kappa_disrupted$. For the 3 diagonal involutions selected by ZS-F5 v1.0 P6 bridge ($\kappa \leq r = 4$): $\kappa_disrupted = 4$ for all three. Hence $I_cell = 4/4 = 1$, and $\delta\varphi_cell = A$ exactly.

[STATUS: DERIVED-under-P6] Conditional on ZS-F5 v1.0 Proposition P6.

5.3 Microscopic Period and Tunneling Action

$T_micro = 2\pi/A \approx 78.45$ [Planck units]. $S_tunnel = (5/2) \times T_micro \approx 196.1$. The factor $5/2 = |I_h|/|O_h|$ is the crystallographic projection cost from Y-sector icosahedral to X-sector octahedral symmetry.

[STATUS: DERIVED-under-P6] S_tunnel from locked constants only.

§6. Z-Anchor Mechanism

The Z-anchor is the topological defect at the center of every Z-Spin vortex ($|\Phi| = 0$), mandated by $\pi_1(U(1)) = \mathbb{Z}$. After $T_micro/\delta\varphi = 1/A \approx 12.5$ oscillations, the accumulated phase drift reaches 2π,

triggering a topological transition (Z-Telomere collapse). U(1) vortex-core mechanism provides robustness: even if the Tolman Z-anchor saturates, the vortex-core pathway remains active.

[STATUS: DERIVED] Topological from $\pi_1(U(1))$ + phase drift.

§7. UV Convergence (Conditional)

The one-loop divergence: $\Delta S_{UV} \sim (A/4\pi)^2 \times \Lambda^4/(M^2_P\, m^2_\varepsilon)$. For $\Lambda = M_P$: $\Delta S_{UV} \sim (A/4\pi)^2 \approx 4.1 \times 10^{-5}$, perturbatively small. Full UV completion requires asymptotic safety embedding.

[STATUS: HYPOTHESIS] UV convergence conditional on asymptotic safety.

§8. RG Fixed Point and λ_vac Determination

8.1 One-Loop β-Function

$$\beta_\lambda = (3/16\pi^2)(\lambda - 6\xi^2)(\lambda - 2\xi^2), \quad \xi = A = 35/437 \qquad (3)$$

[STATUS: DERIVED-CONDITIONAL] Conditional on Elizalde et al. (1995) form.

8.2 Fixed Point Structure

$\lambda_- = 2A^2 = 0.01283$ [IR stable]. $\lambda_+ = 6A^2 = 0.03849$ [IR unstable]. Stability: $d\beta_\lambda/d\lambda$ at $\lambda_- = (3/16\pi^2)(-4A^2) < 0 \rightarrow$ IR stable. Note: $\lambda = A$ and $\lambda = 4A^2$ are NOT fixed points.

[STATUS: PROVEN] Under β-function form (3).

8.3 RG Trajectory: λ_inf → λ_vac

$\lambda_{inf} = 7.63 \times 10^{-12}$ (ZS-U1 v1.0 §4.2) satisfies $\lambda_{inf} \ll \lambda_-$. Therefore $\beta_\lambda(\lambda_{inf}) > 0$: coupling increases toward IR. Unique IR attractor: $\lambda_{vac} = 2A^2$.

8.4 Physical Consequences

$m_\rho = \sqrt{(2\lambda_{vac})} \times M_P = 2A \times M_P = 0.1602 \times M_P$. $\ell_c = \ell_P/\sqrt{\lambda_{vac}} = \ell_P/(A\sqrt{2}) = 8.83\ \ell_P$. The condition $m_\rho \gg H_0$ is satisfied. Fifth-force suppression holds for any $\lambda_{vac} > 10^{-120}$.

8.5 Comparison with λ_vac = A (Hypothesis)

The alternative $\lambda_vac = A$ is not a β-function fixed point but follows from a different mechanism: the critical curvature self-consistency condition. This remains HYPOTHESIS pending numerical relativity verification. The two candidates differ in kink width: $\ell_c = 3.53\ \ell_P$ ($\lambda=A$) vs $8.83\ \ell_P$ ($\lambda=2A^2$), distinguishable at 2.5× in NR simulations.

§9. Falsification Conditions

Table 2. Falsification gates.

ID	Condition	Timeline	Status
FU5-1	(Z,X,Y) = (2,3,6) contradicted by full LQG	~2030	TESTABLE
FU5-2	BH entropy requires $\gamma \neq \ln 2/(\pi\sqrt{3})$	Ongoing	TESTABLE
FU5-3	$\delta\varphi \neq A$ at 2-loop level	~2028	TESTABLE
FU5-4	Z-anchor not found in BH simulations	~2035	TESTABLE
FU5-5	ZS-F5 v1.0 P6 falsified ($\kappa \neq 4$)	~2028	TESTABLE
FU5-6	β_λ form contradicted at 2-loop (>30%)	Theory	TESTABLE
FU5-7	NR measures $\ell_c = 3.53\ \ell_P$ (not 8.83)	NR sims	TESTABLE
FU5-8	Threshold corrections drive λ past λ_+	Theory	TESTABLE

§10. Non-Claims

NC1: ZS-U5 does NOT predict a specific numerical value of the Immirzi parameter from A. The connection is structural, not numerical.

NC2: ZS-U5 does NOT claim UV finiteness. Only conditional convergence under asymptotic safety.

NC3: ZS-U5 scope is strictly quantum gravity and topological structure. Post-inflationary thermalization, decay channels, and detector predictions belong exclusively to ZS-U2 v1.0.

NC3b: The Regge-Holonomy derivation is conditional on ZS-F5 v1.0 P6 (primitive cell bridge selection).

NC4: $\lambda_vac = 2A^2$ is conditional on the one-loop Elizalde et al. β-function form. Two-loop corrections shift the fixed point by $O(A^4/\pi^2)$ and are left for future work.

§11. Conclusion

ZS-U5 establishes the structural bridge between Z-Spin Cosmology and loop quantum gravity. The Z-Spin Q=11 slot register and the LQG 4-valent quantum tetrahedron share a structural correspondence organized by the tetrahedral symmetry group T_d. The group-theoretic quantities $|T_d| = 24$,

$|Stab_{T_d}(v)| = 6$, and $|I_h/T_d| = 5$ are rigorously proven and determine all Z-Spin timescales independently of the correspondence hypothesis.

Explicit non-identity: $Q = 11 \neq \dim(\text{intertwiner}) = 2 \neq \dim(\text{Hilbert}) = 16$. The Z-Spin register counts sector slots from polyhedral geometry (ZS-F2 v1.0), not quantum states from SU(2) recoupling.

The Z-Telomere mechanism provides dynamical content: phase drift $\delta\varphi = A$ (DERIVED-under-P6), microscopic period $T_{micro} = 2\pi/A$, and tunneling action $S_{tunnel} = (5/2)T_{micro}$, all from locked constants with zero new parameters. The one-loop RG analysis (§8) determines $\lambda_{vac} = 2A^2 = 0.01283$, completing the ZS-U5 scope.

§12. Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/NumPy) is publicly available.

Appendix A. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Locked Inputs	5	5/0	A, Q, (Z,X,Y), κ=4, γ_LQG
[B] Structural Correspondence	8	8/0	SU(2) dim=2/16, Q=11(ZS-F2), T_d shared
[C] Z-Telomere & Phase Drift	5	5/0	δφ=A, T_micro=78.45, S_tunnel=196.1
[D] RG Fixed Points	6	6/0	λ_vac=2A², stability, β_λ zeros
[E] Physical Consequences	5	5/0	m_ρ=2AM_P, ℓ_c=8.83ℓ_P
[F] Cross-Paper Consistency	5	5/0	Dependency chain, interface table

TOTAL: 34/34 PASS — 100% pass rate

Interface Table

Interface	Value	Source/Destination
A = 35/437	Locked	ZS-F2 v1.0 → ZS-U5
Q=11, (Z,X,Y)=(2,3,6)	Locked	ZS-F2 v1.0 → ZS-U5
κ = 4	Locked	ZS-F5 v1.0 P6 → ZS-U5
γ_LQG = ln2/(π√3)	Standard	LQG literature → ZS-U5
S_tunnel = 196.1	Output	ZS-U5 → ZS-A3 v1.0
λ_vac = 2A²	Output	ZS-U5 §8 → ZS-F1 v1.0 §4.4

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Version History

v1.0 (March 2026): Initial public release. Structural correspondence Q=11 ↔ quantum tetrahedron (§2, HYPOTHESIS — explicit non-identity: Q=11 ≠ dim(intertwiner)=2 ≠ dim(Hilbert)=16). BH entropy ln 2 origin. Immirzi parameter connection (two paths). Z-Telomere collapse (δφ = A, DERIVED-under-P6). Regge-Holonomy framework (§5.2). Z-anchor mechanism. UV convergence (conditional). One-loop RG trajectory (§8: $\lambda_{vac} = 2A^2$, DERIVED-CONDITIONAL). 8 falsification gates (FU5-1–FU5-8). Interface table. ZS-M3 v1.0 added to references. 34/34 tests. (Consolidated from internal research notes up to v4.0.0)